

STRINGS AND FILE I/O

Aim:

To write and execute a Python program to manipulate strings and perform file input and output operations.

Interactive Development Environment (IDE) Used :

Jupyter NoteBook

Programming Language Used :

Python

Python Concepts used:

String methods are used to manipulate and process text data. File I/O (Input/Output) is used to create, write, read and process data stored in external files. In this experiment, string methods, text file handling and CSV file handling are demonstrated.

Concept	Example Code	Description
String Creation	<code>name = "Gear"</code>	Creates a string using single or double quotation marks.
String Indexing	<code>name[0]</code>	Accesses an individual character using its index. Indexing starts from 0.
Negative Indexing	<code>name[-1]</code>	Accesses characters from the end of the string. -1 represents the last character.
String Slicing	<code>name[1:4]</code>	Extracts a portion of a string using start and stop indexes; the stop index is excluded.
String Concatenation	<code>a + b</code>	Joins two strings together.
String Repetition	<code>name * 3</code>	Repeats a string the specified number of times.
String Length	<code>len(name)</code>	Returns the number of characters in the string.
Uppercase	<code>name.upper()</code>	Converts alphabetic characters to uppercase.
Lowercase	<code>name.lower()</code>	Converts alphabetic characters to lowercase.
Remove Spaces	<code>name.strip()</code>	Removes leading and trailing whitespace.
Remove Left Spaces	<code>name.lstrip()</code>	Removes whitespace from the beginning of the string.
Remove Right Spaces	<code>name.rstrip()</code>	Removes whitespace from the end of the string.
Replace Text	<code>name.replace("gear", "shaft")</code>	Replaces specified text with another text.
Count Occurrences	<code>name.count("a")</code>	Counts occurrences of a character or substring.
Find Position	<code>name.find("ear")</code>	Returns the index of the first occurrence; returns -1 if not found.
Check Starting Text	<code>name.startswith("ge")</code>	Checks whether the string starts with the specified text.
Check Ending Text	<code>name.endswith("ar")</code>	Checks whether the string ends with the specified text.
Split String	<code>name.split()</code>	Splits a string into a list of parts.
Join Strings	<code>"-".join(parts)</code>	Joins elements of a sequence using the specified separator.
Remove Outer Characters	<code>name.strip(" ")</code>	Removes specified characters from both ends.
String Membership	<code>"gear" in name</code>	Checks whether a substring is present in a string.
String Comparison	<code>name == "Gear"</code>	Compares two strings for equality.
Open File for Writing	<code>open("data.txt", "w")</code>	Opens a file in write mode. Creates the file if it does not exist; existing contents are replaced.
Open File for Reading	<code>open("data.txt", "r")</code>	Opens an existing file in read mode.
Open File for Appending	<code>open("data.txt", "a")</code>	Opens a file in append mode. New data is added at the end without removing existing data.

Open File for Read/Write	<code>open("data.txt", "r+")</code>	Opens an existing file for both reading and writing.
Write to File	<code>file.write("Gear\n")</code>	Writes a string to a text file.
Write Multiple Lines	<code>file.writelines(lines)</code>	Writes multiple strings to a file. Newline characters must be included when required.
Read Entire File	<code>file.read()</code>	Reads the complete remaining contents of a file.
CSV Module	<code>import csv</code>	Imports Python's CSV module for working with comma-separated data.
CSV Writer	<code>csv.writer(file)</code>	Creates a writer object for writing CSV data.
CSV Reader	<code>csv.reader(file)</code>	Creates a reader object for reading CSV rows.

Python Program:

```
import csv
# 1. String methods
name = " gear "
print("Original:", name)
print("Uppercase:", name.upper())
print("Lowercase:", name.lower())
print("Without spaces:", name.strip())
print("Length:", len(name.strip()))
print("Replaced:", name.replace("gear", "shaft"))
print("Count of 'a':", name.count("a"))
# 2. Write to a text file
with open("data.txt", "w") as file:
    file.write("Gear\n")
    file.write("Steel\n")
    file.write("120 mm")
# 3. Read from text file
with open("data.txt", "r") as file:
    print("\nText file data:")
    print(file.read())
# 4. Write to CSV file
with open("data.csv", "w", newline="") as file:
    writer = csv.writer(file)
    writer.writerow(["Name", "Material", "Diameter"])
    writer.writerow(["Gear", "Steel", "120 mm"])
print("\nCSV file created successfully.")
```

Sample Output :

Result :

The Python program was written and executed successfully to demonstrate string manipulation, text file reading and writing, and CSV file creation.